

BIOLOGY EXAM-18 : DETAILED MARKING REPORT

Chapters 1 to 7 (revision of the whole course studied so far)

Seemab · Class 9 · Attempt Date: 2026-09-24 · Paper: 50 marks, 2h15m, no optional questions

Final Score: 48 / 50 · 96% · Grade A

Marking notes. All three sections (MCQ list, Section B, Section C) are present and in full sequence; the 7-page CamScanner scan is clearly legible throughout (110 to 600 DPI crops used on every equation, label and the MCQ answer list). She wrote her MCQ answers as a plain numbered list on the last page (page 7), per her usual habit, not in a bubble grid. A blind second marker independently worked the same script and confirmed 48/50: no mark moves, every section and per-chapter row and the deduction log agree, and no key errors were found. One naming item (Q.7ii, "link reaction") is recorded below as a resolved judgement call, credited in full. This is a straight revision paper on already-taught content, so the score is not split into new-chapter/revision, per the paper's own design.

9 / 10 SECTION A (MCQS)		23.5 / 24 SECTION B (SHORT)		15.5 / 16 SECTION C (LONG)		
Ch1 5/5	Ch2 5/5	Ch3 5/5	Ch4 9/9	Ch5 5/6	Ch6 6/6	Ch7 13/14

SECTION A : Multiple Choice Questions (9 / 10)

#	Question (summary)	Correct	Seemab's	Result
1	Ch1: inserting human insulin gene into bacteria	B : biotechnology	B	1
2	Ch2: colonial organization is unique to	C : Protista	C	1
3	Ch3: correct statement about ribosomes	A	A	1
4	Ch4: what halves chromosome number in meiosis	D : homologous separation, meiosis I	D	1
5	Ch5: Table 5.2, removes pathogens, fights infection, heals	B : immune system	A : lymphatic system	0
6	Ch5: structure at a different level of organization	A : neuron	A	1
7	Ch6: disaccharide also called the transport sugar	D : sucrose	D	1
8	Ch6: all RNA nucleotides differ from DNA in having a different	C : pentose sugar	C	1
9	Ch7: ATP wavy-line bonds are called	A : high energy bonds	A	1
10	Ch7: products of the light reactions	B : ATP, NADPH and oxygen	B	1

Only MCQ 5 wrong. This is the exact confusable pair the paper deliberately re-tested from exam-16 Q12 (lymphatic vs immune system, Table 5.2), and she chose the same wrong option again : see the strengths/weaknesses note below. MCQ 8 was flagged before sending as the item most likely to be contested (a pencil mark against "nitrogen base" in her own book copy); she answered it correctly as pentose sugar, so that risk did not materialise.

SECTION B : Short Questions (23.5 / 24)

Q.2 (Chapter 1) : The spread of malaria 4 / 4

(i) "It was hypothesized that mosquitoes spread malaria from one person to another." 1/1

(ii) "Ronald Ross allowed mosquitoes to feed on infected sparrows and then allowed those mosquitoes to bite healthy sparrows. The healthy sparrows developed malaria, proving that mosquitoes could transmit the disease." 2/2

(iii) "The hypothesis was tested by allowing mosquitoes that had fed on malaria patients to bite healthy people. The people then developed malaria, confirming mosquito transmission." 1/1

Full marks. Part (iii) does not name the "Italian biologist" or "Anopheles" as the key's worked answer does, but the question itself never asks for either name, and the causal chain she gives (mosquito bites patient, then bites healthy person, person develops malaria, confirming transmission) is complete and correct.

Q.3 (Chapter 2) : Differences 4 / 4

(i) Bacteria vs Protists: "They are prokaryotic cells and lack a nucleus" / "They are eukaryotic cells and have a nucleus" ; "They generally lack membrane bound organelles" / "They possess membrane bound organelles" 2/2

(ii) Fungi vs Plants: "It cannot make its own food and obtain nutrients from other sources" / "It makes its own food by process called photosynthesis" ; "Their cell walls are mainly made of chitin" / "Their cell walls are mainly made of cellulose" 2/2

All four differences correct and each pair is genuinely two-sided (both columns filled, not one column copied from the other).

Q.4 (Chapter 3) : SER and the salivary gland cell4 / 4

(i) "Functions:- 1) Lipid synthesis 2) Detonification of drugs & harmful substances."

2/2

(ii) "Salivary gland cell:- Two abundant organelles are:- Ribosomes - for making protions/enzymes. RER (rough endaplasmic reticulum) - for synthesis and transport of proteins"

2/2

Both organelles named with a sound reason each, matching the key's accepted list (ribosomes/RER for protein synthesis). "Detonification" is a spelling slip for "detoxification"; the meaning is unambiguous, so no mark lost.

Q.5 (Chapter 5) : Plant tissues4 / 4

(i) Xylem vs Phloem : "It transports water and mineral salts from roots to other parts" / "It transports food/sugars from source to sinks" ; "Mainly consists of dead cell" / "Mainly consists of living cell"

2/2

(ii) Dermal vs Ground tissue : "Formes the outer protective covering of the plant" / "Makes up the tissue b/w dermal and vascular tissues" ; "Protects against physical damage and water loss" / "Performs functions such as photosynthesis, storage and support"

2/2

Full marks, book-accurate wording throughout.

Q.6 (Chapter 6) : Carbohydrates and chromatin4 / 4

(i) "Carbon : Hydrogen : Oxygen = 1 : 2 : 1. General formula:- (CH2O)n"

2/2

(ii) "Chromatin:- is composed of DNA & proteins, mainly histones. When DNA coils around histones protein, [word struck out] it forms bead-like structure called nucleosomes."

2/2

Ratio, general formula, chromatin composition and nucleosomes all correct.

Q.7 (Chapter 7) : Respiration3.5 / 4

(i) "Glycosis take places in cytoplasm. One glucose molecule is converted into 2 pyruvic acid molecules, producing a net gain of 2ATP. Equation:- Glucose → 2 Pyruvic acid + 2ATP"

2/2

(ii) "In order:- Glycosis, Link reaction, Krebs cycle, Electron transport chain."

(nothing further written about where oxygen is used)

1.5/2

All four stages named in the correct order, so the 1.5 marks for the sequence are credited in full : the question asks for the stages "in the order given in your textbook", constraining the order, not the vocabulary, and the key allocates these 1.5 marks for correct order alone (see the judgement call below on "Link reaction"). But the question also explicitly asks her to state the stage in which oxygen is used, worth 0.5 mark on its own, and that sentence is never written : nothing follows the four names. Half mark lost for that specific missing part, not for the naming.

SECTION C : Long Questions (15.5 / 16)

Q.8 (Chapter 7) : Photosynthesis7.5 / 8

(i) Labels : "A : H2O B : CO2 C : O2 D : Sugar"

4/4

(ii) Balanced equation : "6CO2 + 6H2O → C6H12O6 + 6O2"

with Light energy written above the arrow and Chlorophyll below it

2/2

All four labels correct. The equation is exactly the shorter Chapter 6 form the answer key explicitly accepts (book p83), coefficients and subscripts all verified at 600 DPI, with both arrow labels present in the right position.

(iii) Two energy-rich molecules : "1. ATP 2. NADP."

1.5/2

ATP correct and full. Her second molecule reads clearly as NADP at 400 to 600 DPI (the third letter's bowl closes back to the baseline, matching the D of her MCQ 4 and MCQ 7 answers; the fourth letter's loop closes at mid-height, matching the P of "ATP" on the line above). The H of NADPH is missing, so it is priced as a missing name at the standard half mark. The concept (the second light-reaction product handed to the Calvin cycle) is clearly understood; only the final letter is off.

Q.9 (Chapter 4) : Meiosis8 / 8

(i) Four differences, meiosis I vs II : "Homologous chromosomes separate" / "Sister chromatids sperate" ; "Produces 2 haploid cells" / "Produces four haploid cells in total" ; "Crossing over occurs in prophase I" / "Crossing over does not occurs" ; "Chromosome number is reduced by half" / "Chromosomes n.o remains the same"

4/4

(ii) 18-chromosome cell : "(a) = 9 chromosome (b) 9 (c) 4"

3/3

(iii) "Bcz crossing over and independent [assortment] produce different combinations of genes/chromosome in the daughter cells."

1/1

Full marks. All four differences are genuine, correctly paired opposites; both numeric parts and the reasoning in (iii) are exactly right. Her strongest question on the paper. Worth flagging by itself; her right-hand column header originally read "Mitosis", which she struck through and corrected to "Meosis II" (her spelling, with "II" added above and underlined) before writing a single one of the four differences. That is the exact reflex that cost her exam-17 Q.3(ii)(c) (comparing mitosis to the whole of meiosis); catching and correcting it herself, unprompted, is real progress, and every difference she goes on to give is a genuine meiosis I vs meiosis II pair.

RE-TEST OUTCOME (deliberately re-served from earlier papers)

Re-test	Result	Evidence
MCQ 5 : lymphatic vs immune system, Table 5.2 (missed as exam-16 Q12)	NOT FIXED	Answered "A" (lymphatic system) again; correct is "B" (immune system). Same confusable pair, same wrong choice, second time.
MCQ 4 : meiosis I mechanism, homologous chromosomes separate (also flagged as a re-test of the exam-17 Q.3(ii)(c) mitosis/meiosis mix-up)	FIXED	Answered "D" (separation of homologous chromosomes during meiosis I), correct.

MCQ 9 : ATP high-energy bonds (missed as exam-16 Q17, hydrogen bonds)	FIXED	Answered "A" (high energy bonds), correct.
Q.9(i) : comparing the right two things in a difference table (missed as exam-17 Q.3(ii)(c), meiosis vs whole mitosis)	FIXED	All four meiosis I vs II differences are genuine, correctly paired; she also caught and struck through her own "Mitosis" header slip before answering, replacing it with "Meiosis II".
Q.8(ii) : photosynthesis equation coefficients (missed as exam-17 Q.2(i) (a), "6C6H12O6")	FIXED	"C6H12O6" written correctly, no stray leading coefficient on the sugar.
MCQ 8 : pentose sugar vs nitrogen base (flagged risk from a pencil mark in her own book)	FIXED	Answered "C" (pentose sugar), correct; the flagged risk did not materialise.

JUDGEMENT CALLS

#	Item	Awarded	Note
1	Q.7(ii) : "Link reaction" used for the book's "Formation of Acetyl CoA"	1.5/2 for order+names, 0 of the 0.5 for the oxygen stage	Credited in full for order/naming: the question constrains the ORDER ("in the order given in your textbook"), not the vocabulary, and the key allocates these 1.5 marks for correct order alone. Penalising the non-book term would be charging her for imprecision in our own wording, not a misconception on her part.

Confirmed by a blind second marking: no open range. **48/50 (96%) is the final mark.** The only genuine losses are MCQ 5 and the 0.5 for the missing oxygen-stage sentence in Q.7(ii).

STRENGTHS AND WEAKNESSES

Strengths: Her best Biology score since exam-13 (99.2%, 31 July) and a clear step up from exam-16 (94.4%) and exam-17 (94.7%). Three of four documented re-set weaknesses landed correctly this time: ATP high-energy bonds (MCQ 9), photosynthesis equation coefficients (Q.8ii), and comparing the right two things in a meiosis difference table (Q.9i), which is also confirmed by MCQ 4 landing correctly (the same exam-17 Q.3(ii)(c) mix-up, re-tested a second way). Q.9 (meiosis, 8/8) is flawless, and she caught her own "Mitosis" header slip mid-answer, struck it through and corrected it to "Meiosis II" before writing a single difference : real, visible self-correction of the exact habit that cost her marks last time. Every difference-table pair across the paper (Q.3, Q.5, Q.9i) is genuinely two-sided with both columns correctly filled, and the photosynthesis figure labels (Q.8i) are all four correct. The missing-conclusion habit, a tracked weakness across many earlier papers, **did not appear on this paper**: Q.2(ii), Q.2(iii) and Q.9(iii) all close on a proper concluding sentence.

Weaknesses: (1) **Lymphatic vs immune system (Table 5.2)** : wrong on MCQ 5 for the second time running (also missed on exam-16 Q12); this is now a confirmed recurring confusable pair, not a one-off. (2) **Answering only part of a multi-part question** : Q.7(ii) named all four respiration stages correctly but never went on to state where oxygen is used, even though the question asks for it explicitly. (3) **NADPH left incomplete** : she wrote NADP in Q.8(iii), correct as far as it goes, but the final H (making it NADPH, the actual light-reaction product) was never added.

COMPARISON : RECENT BIOLOGY ATTEMPTS

Exam	Date	Score	%	MCQ	Note
exam-15-ch1-7	21 Aug 2026	39/43	90.7%	:	
exam-16-ch1-7	13 Sep 2026	34/36	94.4%	18/20	Lymphatic vs immune first missed here (Q12)
exam-17-ch1-7 (Version B)	17 Sep 2026	36/38	94.7%	10/10	Perfect MCQs, coefficient and difference-table slips
exam-18-ch1-7	24 Sep 2026	48/50	96%	9/10	Best Biology score since exam-13 (99.2%); lymphatic/immune pair still unresolved

A genuine, sustained improvement across three consecutive papers (94.4% → 94.7% → 96%), with three of four targeted re-tests fixed this time. The one repeat miss (MCQ 5, lymphatic vs immune) is now a confirmed two-time confusable pair and is the clearest single revision target from this paper.

: End of Report :